10.1.2 Human alpha lactalbumin

Alpha lactalbumin is a protein composed of 123 amino acids and 4 disulphide bridges, with a molecular weight of 14178 Da. In human milk, α lactalbumin is the most abundant protein comprising 25% of total protein found in human milk. It is synthesized by the mammary glands. α lactalbumin binds calcium and zinc ions and possesses bactericidal and anti tumour activities.

Improvement of the nutritional value of cow's milk with transgenic expression of recombinant human alpha lactalbumin has been attempted. Healthy transgenic cows were produced by somatic nuclear transfer, in which expression of upto 1.55g/L of recombinant human alpha lactalbumin was achieved. Similarly transgenic goats were also produced, in which the expression of Human alpha lactalbumin was about 0.1 to 0.9mg/mL.

Somatic cell nuclear transfer is a technique for creating a viable embryo from a body cell and an egg cell. This technique is discussed later in animal cloning.



In 1997, Rosie, the first transgenic cow produced human protein enriched milk, which contained the human alpha lactalbumin. The protein rich milk (2.4 gm/litre) was a nutritionally balanced food for new born babies than the normal milk produced by the cows.

10.1.3 Interferons

Interferons are proteinaceous, antiviral, species specific substances produced by mammalian cells when infected with viruses. Interferons were discovered by Alick Isaacs and Jean Lindemann in 1957. Based on the structure of interferons they are classified as α, β and γ interferons. They stimulate the cellular DNA to produce antiviral enzymes which inhibit viral replication and protect the cells. Interferons could be isolated from blood, but the amount of blood required for isolation of interferons is enormous and not practical. To overcome this issue interferons could be produced by rDNA technology. The yeast *Saccharomyces cerevisiae* is more suitable for production of recombinant interferons than *E.coli*, since *E.coli* does not possess the machinery for glycosylation of proteins. Interferons are used for the treatment of various diseases like cancer, AIDS, multiple sclerosis, hepatitis C and herpes zoster. In spite of the therapeutic applications interferons are not within the reach of the common man due to high cost for its production.

10.1.4 Recombinant Vaccines

Recombinant DNA technology has been used to produce new generation vaccines. The limitations of traditional vaccine production could be overcome by this approach.

The recombinant vaccines are generally of uniform quality and produce less side effects as compared to the vaccines produced by conventional methods. Different types of recombinant vaccines include subunit recombinant vaccines, attenuated recombinant vaccines and DNA vaccines.

Subunit recombinant vaccines

Vaccines that use components of a pathogenic organism rather than the whole organism are called **subunit vaccines**; recombinant DNA technology is very suited for developing new subunit vaccines. It includes components like proteins, peptides and DNAs of pathogenic organisms. The advantages of these vaccines include their purity in preparation, stability and safe use.

Attenuated recombinant vaccines

This includes genetically modified pathogenic organisms (bacteria or viruses) that are made nonpathogenic and are used as vaccines. It is now possible to genetically engineer the organisms (bacteria or viruses) and use them as live vaccines and such vaccines are referred to as attenuated recombinant vaccines.

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